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EXP 16: FOOTPRINT A WIRELESS NETWORK

AIM :

To perform wireless network footprinting — identifying nearby networks, their SSIDs, channels, encryption type, and connected clients — using Kali Linux tools, in a simulated/offline lab setting.

TOOLS REQUIRED :

- Kali Linux
- Wireshark
- Aircrack-ng suite (airodump-ng, used here to read a saved capture file instead of live capture)
- A sample .pcap wireless capture file (Aircrack-ng ships sample test captures, or your instructor will provide one)

PROCEDURE :

Step 1 — Open the capture file in Wireshark

```
wireshark sample_capture.pcap
```

Step 2 — Filter for beacon frames (this reveals every advertised network)

In the Wireshark filter bar:

```
wlan.fc.type_subtype == 0x08
```

Each result is one Access Point announcing itself. Expand a packet to see the **SSID**, **channel**, and **supported rates**.

Step 3 — Identify encryption type

Look inside the "Tagged Parameters" section of the beacon frame for **RSN Information** (WPA2) or the absence of it (Open network) or **WEP Parameters**.

Step 4 — Identify connected clients

Filter for data frames to see device (client) MAC addresses talking to a specific AP:

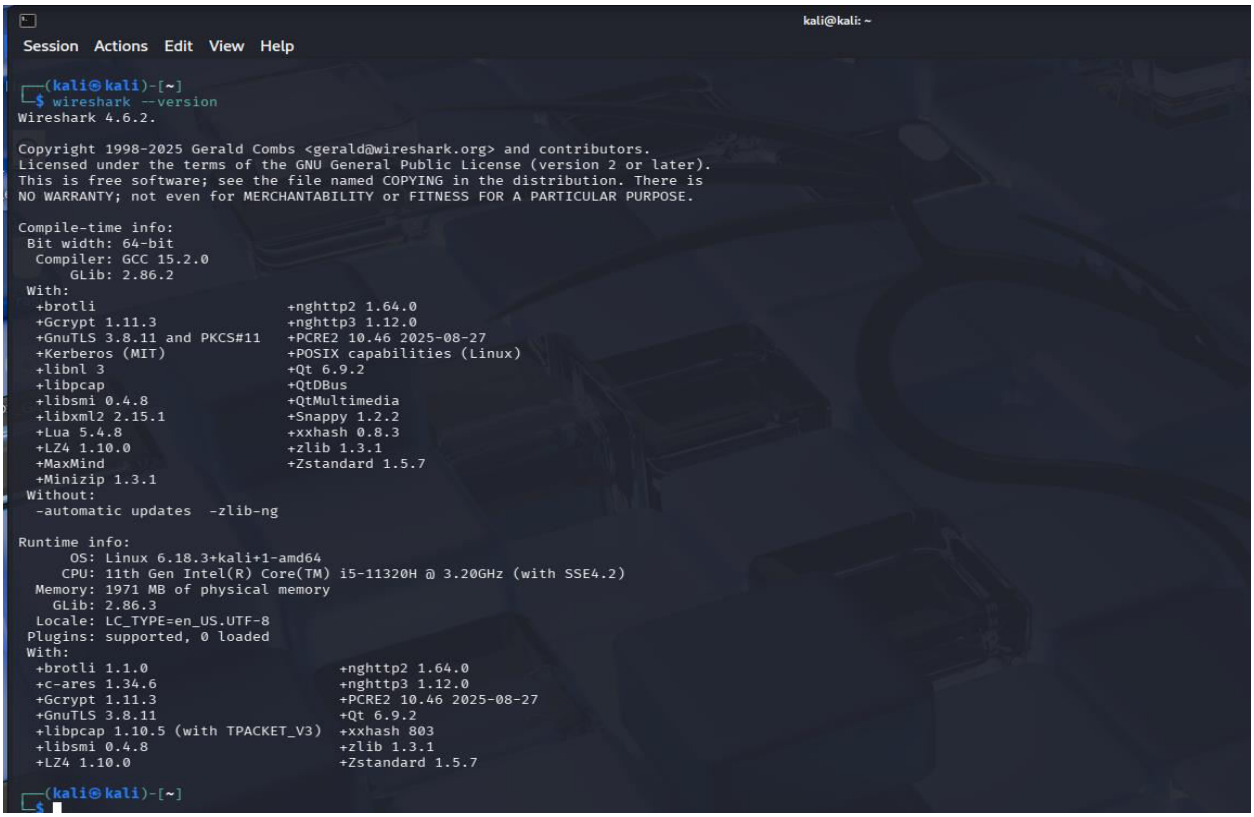
```
wlan.addr == <BSSID-of-target-AP>
```

Step 5 — Analyze the same file with airodump-ng (offline read mode)

```
airodump-ng --read sample_capture.pcap
```

This produces the classic table view: BSSID, PWR, channel, encryption (WEP/WPA/WPA2), ESSID, and associated client stations.

EXPECTED OUTPUT :

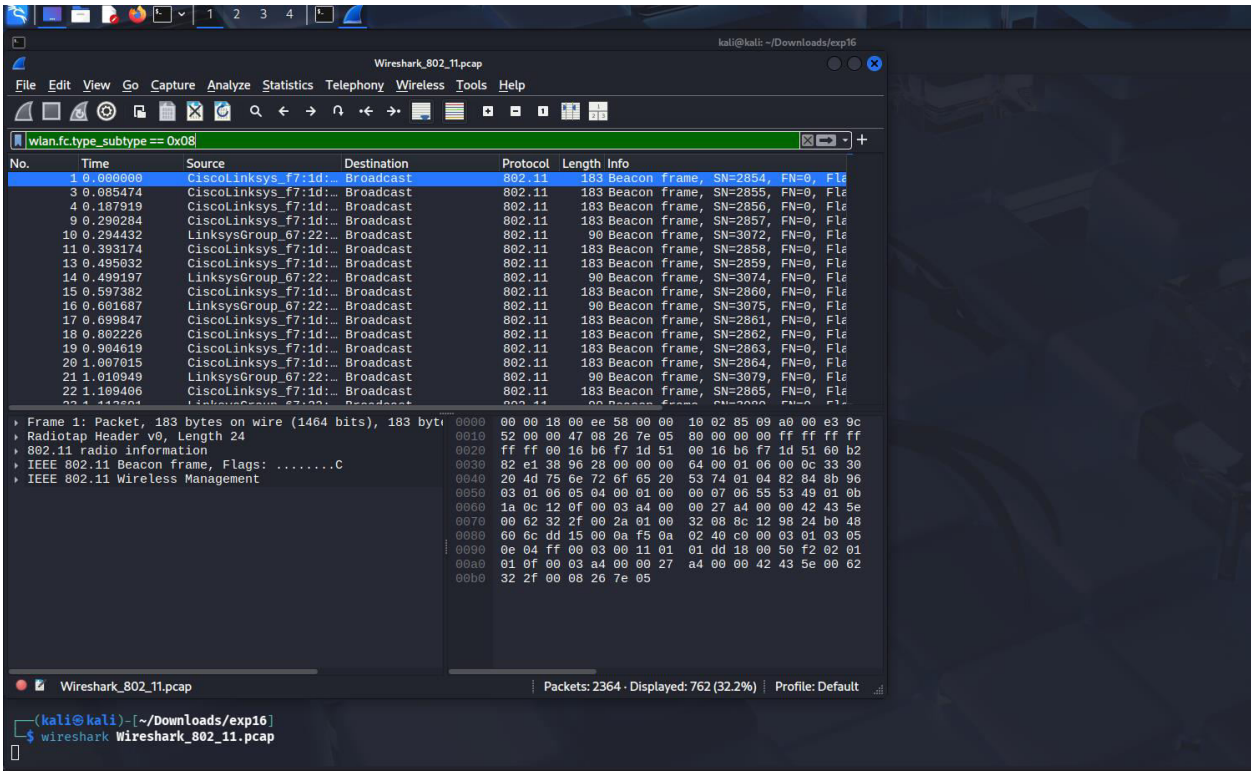


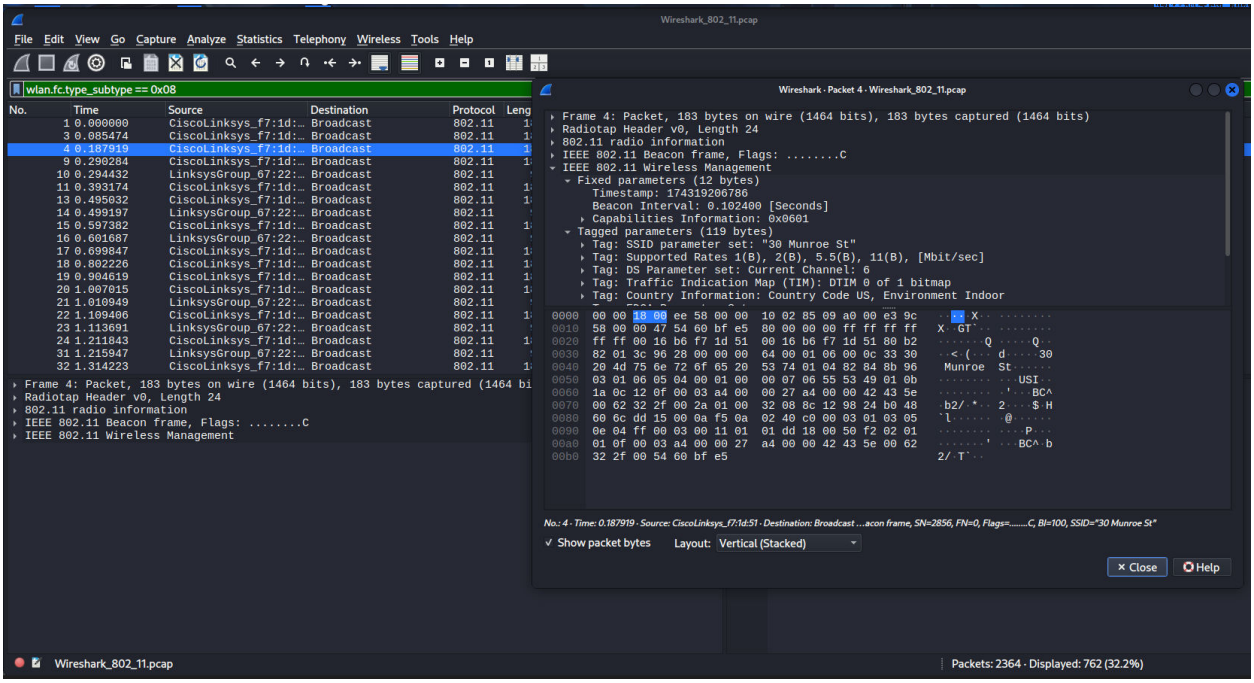


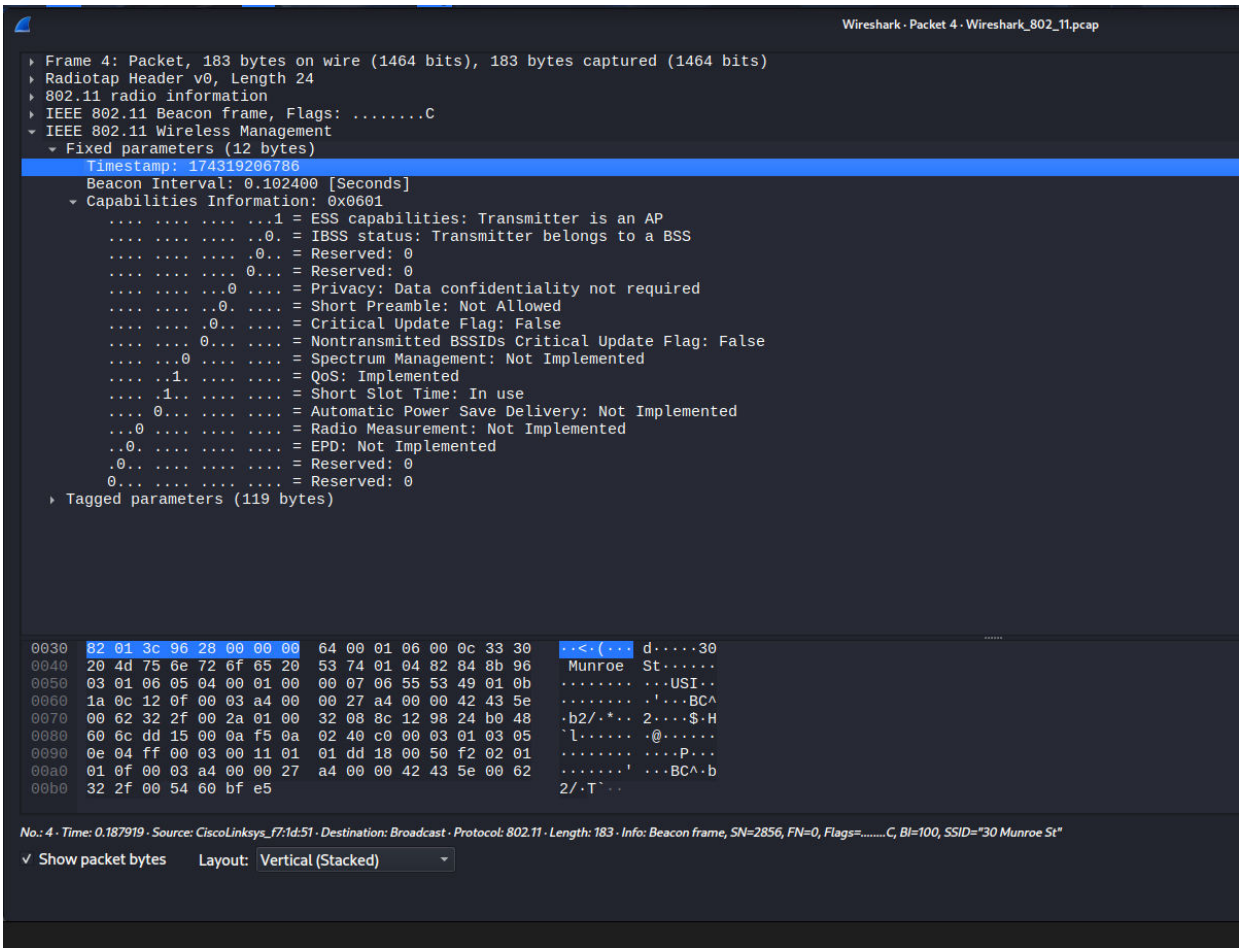
```
┌──(kali㉿kali)-[~/Downloads/exp16]
└─$ capinfos Wireshark_802_11.pcap
File name:           Wireshark_802_11.pcap
File type:           Wireshark/tcpdump/... - pcap
File encapsulation:  IEEE 802.11 plus radiotap radio header
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   2,364
File size:           637 kB
Data size:           599 kB
Capture duration:    73.655470 seconds
Earliest packet time: 2007-06-28 22:05:07.072457
Latest packet time:   2007-06-28 22:06:20.727927
Data byte rate:      8,144 bytes/s
Data bit rate:       65 kbps
Average packet size: 253.75 bytes
Average packet rate: 32 packets/s
SHA256:              37b42baddef3225ca065dd69d5a73b4b8491034aed5de6c52f2f6a319e5a2738
SHA1:                0d442e9e1a6aae57edbd009648f1e4204b12e125
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = IEEE 802.11 plus radiotap radio header (23 - ieee-802-11-radiotap)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 2364
```

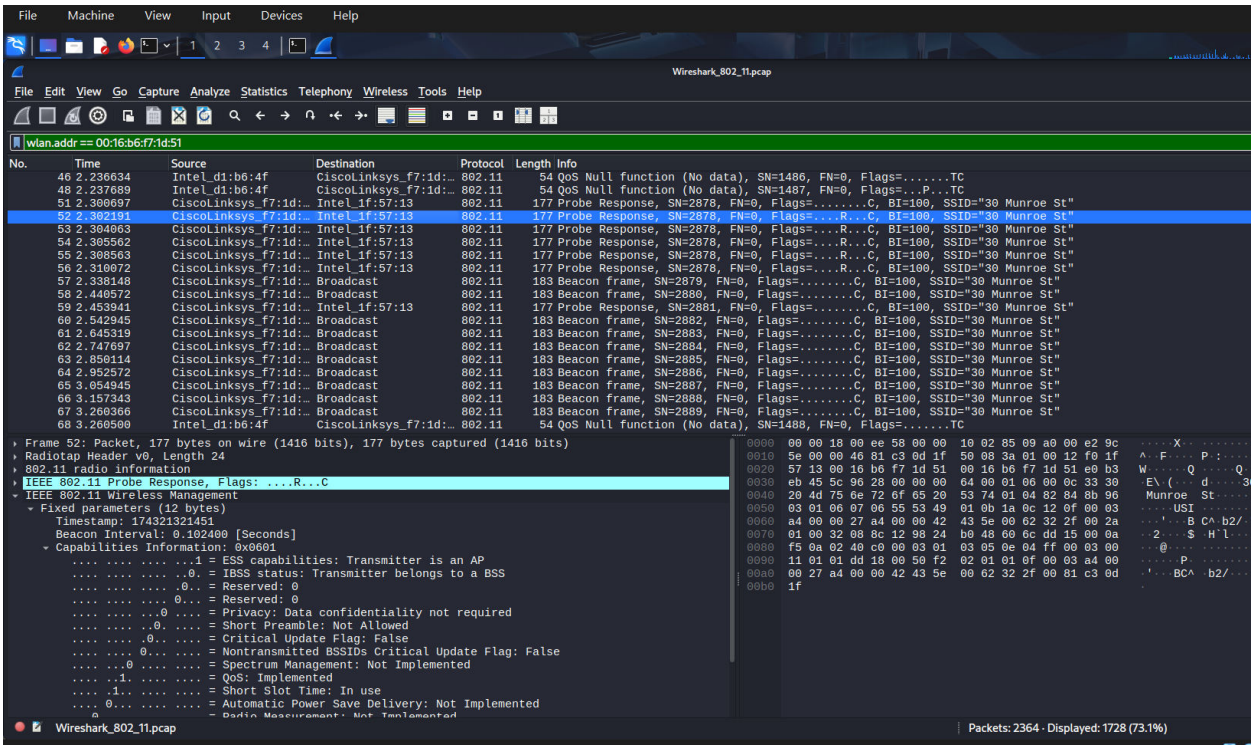
```
┌──(kali㉿kali)-[~/Downloads/exp16]
└─$ tshark -r Wireshark_802_11.pcap \
> -Y "wlan.fc.type_subtype == 0x08" \
> -Y "wlan.fc.type_subtype == 0x08" \
-T fields -e frame.number -e wlan.sa -e wlan.ssid | head -n 15
1       00:16:b6:f7:1d:51       3330204d756e726f65205374
3       00:16:b6:f7:1d:51       3330204d756e726f65205374
4       00:16:b6:f7:1d:51       3330204d756e726f65205374
9       00:16:b6:f7:1d:51       3330204d756e726f65205374
10      00:06:25:67:22:94       6c69ee0104e2273a32
11      00:16:b6:f7:1d:51       3330204d756e726f65205374
13      00:16:b6:f7:1d:51       3330204d756e726f65205374
14      00:06:25:67:22:94       6c696e6b7379733132
15      00:16:b6:f7:1d:51       3330204d756e726f65205374
16      00:06:25:67:22:94       6c696e6b7379733132
17      00:16:b6:f7:1d:51       3330204d756e726f65205374
18      00:16:b6:f7:1d:51       3330204d756e726f65205374
19      00:16:b6:f7:1d:51       3330204d756e726f65205374
20      00:16:b6:f7:1d:51       3330204d756e726f65205374
21      00:06:25:67:22:94       6c696e6b7379733132

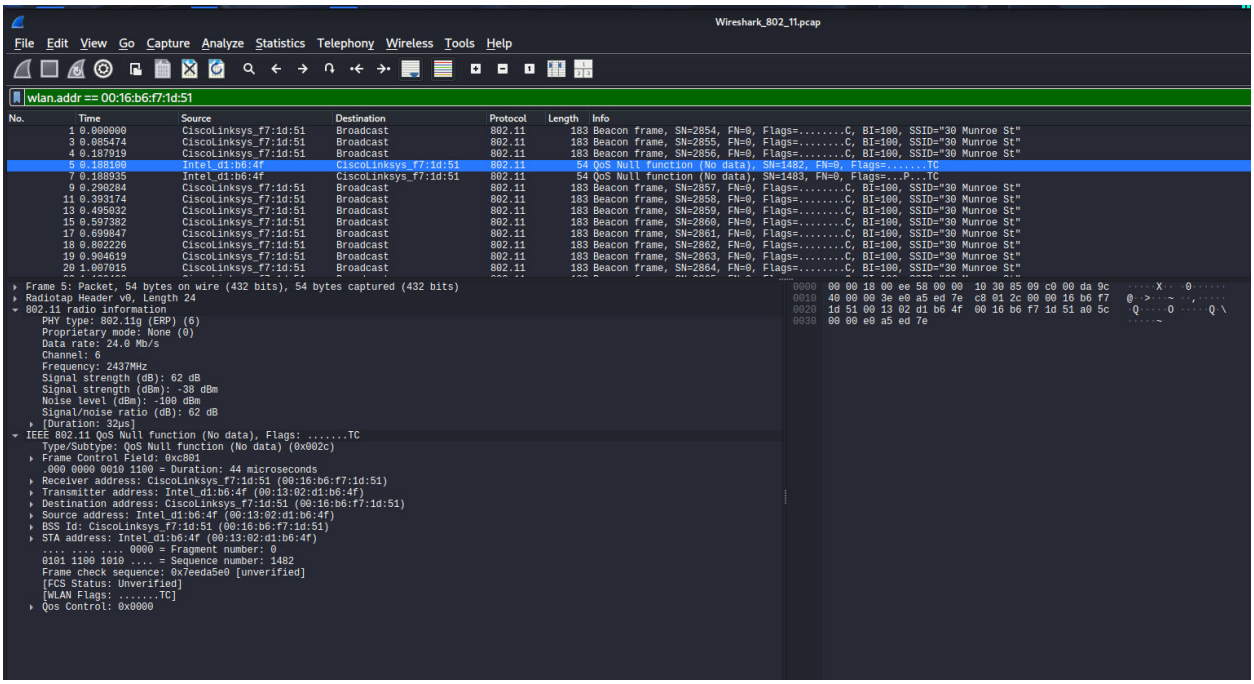
┌──(kali㉿kali)-[~/Downloads/exp16]
└─$
```


kali@kali: ~/Downloads/exp16
Wireshark_802_11.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
wlan.fc.type_subtype == 0x08
No. Time Source Destination Protocol Length Info
1 0.000000 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2854, FN=0, Fla
3 0.085474 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2855, FN=0, Fla
4 0.187919 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2856, FN=0, Fla
9 0.290284 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2857, FN=0, Fla
10 0.294432 LinksysGroup_67:22:... Broadcast 802.11 90 Beacon frame, SN=3072, FN=0, Fla
11 0.393174 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2858, FN=0, Fla
13 0.495032 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2859, FN=0, Fla
14 0.499197 LinksysGroup_67:22:... Broadcast 802.11 90 Beacon frame, SN=3074, FN=0, Fla
15 0.597382 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2860, FN=0, Fla
16 0.601687 LinksysGroup_67:22:... Broadcast 802.11 90 Beacon frame, SN=3075, FN=0, Fla
17 0.699847 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2861, FN=0, Fla
18 0.802226 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2862, FN=0, Fla
19 0.904619 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2863, FN=0, Fla
20 1.007015 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2864, FN=0, Fla
21 1.010949 LinksysGroup_67:22:... Broadcast 802.11 90 Beacon frame, SN=3079, FN=0, Fla
22 1.109406 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2865, FN=0, Fla
Frame 1: Packet, 183 bytes on wire (1464 bits), 183 byt
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 Beacon frame, Flags: C
IEEE 802.11 Wireless Management
0000 00 00 18 00 ee 58 00 00 10 02 85 09 a0 00 e3 9c
0010 52 00 00 47 08 26 7e 05 80 00 00 00 ff ff ff ff
0020 ff ff 00 16 b6 f7 1d 51 00 16 b6 f7 1d 51 60 b2
0030 82 e1 38 96 28 00 00 00 64 00 01 06 00 0c 33 30
0040 20 4d 75 6e 72 6f 65 20 53 74 01 04 82 84 8b 96
0050 03 01 06 05 04 00 01 00 00 07 06 55 53 49 01 0b
0060 1a 0c 12 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e
0070 00 62 32 2f 00 2a 01 00 32 08 8c 12 98 24 b0 48
0080 60 6c dd 15 00 0a f5 0a 02 40 c0 00 03 01 03 05
0090 0e 04 ff 00 03 00 11 01 01 dd 18 00 50 f2 02 01
00a0 01 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e 00 62
00b0 32 2f 00 08 26 7e 05
Wireshark_802_11.pcap
Packets: 2364 · Displayed: 762 (32.2%)
Profile: Default
(kali㉿kali)-[~/Downloads/exp16]
$ wireshark Wireshark_802_11.pcap


Wireshark_802_11.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
wlan.fc.type_subtype == 0x08
No. Time Source Destination Protocol
1 0.000000 CiscoLinksys_f7:1d:... Broadcast 802.11
3 0.085474 CiscoLinksys_f7:1d:... Broadcast 802.11
4 0.187919 CiscoLinksys_f7:1d:... Broadcast 802.11
9 0.290284 CiscoLinksys_f7:1d:... Broadcast 802.11
10 0.294432 LinksysGroup_67:22:... Broadcast 802.11
11 0.393174 CiscoLinksys_f7:1d:... Broadcast 802.11
13 0.495032 CiscoLinksys_f7:1d:... Broadcast 802.11
14 0.499197 LinksysGroup_67:22:... Broadcast 802.11
15 0.597382 CiscoLinksys_f7:1d:... Broadcast 802.11
16 0.601687 LinksysGroup_67:22:... Broadcast 802.11
17 0.699847 CiscoLinksys_f7:1d:... Broadcast 802.11
18 0.802226 CiscoLinksys_f7:1d:... Broadcast 802.11
19 0.904619 CiscoLinksys_f7:1d:... Broadcast 802.11
20 1.007015 CiscoLinksys_f7:1d:... Broadcast 802.11
21 1.010949 LinksysGroup_67:22:... Broadcast 802.11
22 1.109406 CiscoLinksys_f7:1d:... Broadcast 802.11
23 1.113691 LinksysGroup_67:22:... Broadcast 802.11
24 1.211843 CiscoLinksys_f7:1d:... Broadcast 802.11
31 1.215947 LinksysGroup_67:22:... Broadcast 802.11
32 1.314223 CiscoLinksys_f7:1d:... Broadcast 802.11
Frame 4: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bi
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 Beacon frame, Flags: C
IEEE 802.11 Wireless Management
Wireshark · Packet 4 · Wireshark_802_11.pcap
Frame 4: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 Beacon frame, Flags: C
IEEE 802.11 Wireless Management
Fixed parameters (12 bytes)
Timestamp: 174319206786
Beacon Interval: 0.102400 [Seconds]
Capabilities Information: 0x0601
Tagged parameters (119 bytes)
Tag: SSID parameter set: "30 Munroe St"
Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
Tag: DS Parameter set: Current Channel: 6
Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
Tag: Country Information: Country Code US, Environment Indoor
0000 00 00 18 00 ee 58 00 00 10 02 85 09 a0 00 e3 9c
0010 58 00 00 47 54 60 bf e5 80 00 00 00 ff ff ff ff
0020 ff ff 00 16 b6 f7 1d 51 00 16 b6 f7 1d 51 80 b2
0030 82 01 3c 96 28 00 00 00 64 00 01 06 00 0c 33 30
0040 20 4d 75 6e 72 6f 65 20 53 74 01 04 82 84 8b 96
0050 03 01 06 05 04 00 01 00 00 07 06 55 53 49 01 0b
0060 1a 0c 12 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e
0070 00 62 32 2f 00 2a 01 00 32 08 8c 12 98 24 b0 48
0080 60 6c dd 15 00 0a f5 0a 02 40 c0 00 03 01 03 05
0090 0e 04 ff 00 03 00 11 01 01 dd 18 00 50 f2 02 01
00a0 01 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e 00 62
00b0 32 2f 00 54 60 bf e5
No.: 4 · Time: 0.187919 · Source: CiscoLinksys_f7:1d:51 · Destination: Broadcast ...acon frame, SN=2856, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
Show packet bytes
Layout: Vertical (Stacked)
Close
Help
Wireshark_802_11.pcap
Packets: 2364 · Displayed: 762 (32.2%)


Wireshark · Packet 4 · Wireshark_802_11.pcap
Frame 4: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 Beacon frame, Flags: C
IEEE 802.11 Wireless Management
Fixed parameters (12 bytes)
Timestamp: 174319206786
Beacon Interval: 0.102400 [Seconds]
Capabilities Information: 0x0601
....1 = ESS capabilities: Transmitter is an AP
....0. = IBSS status: Transmitter belongs to a BSS
.... .0.. = Reserved: 0
.... 0... = Reserved: 0
....0 .... = Privacy: Data confidentiality not required
....0. .... = Short Preamble: Not Allowed
.... .0.. .... = Critical Update Flag: False
.... 0... .... = Nontransmitted BSSIDs Critical Update Flag: False
....0 .... = Spectrum Management: Not Implemented
.... ..1. .... = QoS: Implemented
.... .1.. .... = Short Slot Time: In use
.... 0... .... = Automatic Power Save Delivery: Not Implemented
...0 .... = Radio Measurement: Not Implemented
..0. .... = EPD: Not Implemented
.0.. .... = Reserved: 0
0... .... = Reserved: 0
Tagged parameters (119 bytes)
0030 82 01 3c 96 28 00 00 00 64 00 01 06 00 0c 33 30
0040 20 4d 75 6e 72 6f 65 20 53 74 01 04 82 84 8b 96
0050 03 01 06 05 04 00 01 00 00 07 06 55 53 49 01 0b
0060 1a 0c 12 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e
0070 00 62 32 2f 00 2a 01 00 32 08 8c 12 98 24 b0 48
0080 60 6c dd 15 00 0a f5 0a 02 40 c0 00 03 01 03 05
0090 0e 04 ff 00 03 00 11 01 01 dd 18 00 50 f2 02 01
00a0 01 0f 00 03 a4 00 00 27 a4 00 00 42 43 5e 00 62
00b0 32 2f 00 54 60 bf e5
No.: 4 · Time: 0.187919 · Source: CiscoLinksys_f7:1d:51 · Destination: Broadcast · Protocol: 802.11 · Length: 183 · Info: Beacon frame, SN=2856, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
Show packet bytes
Layout: Vertical (Stacked)


File Machine View Input Devices Help
Wireshark_802_11.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
wlan.addr == 00:16:b6:f7:1d:51
No. Time Source Destination Protocol Length Info
46 2.236634 Intel_d1:b6:4f CiscoLinksys_f7:1d:... 802.11 54 QoS Null function (No data), SN=1486, FN=0, Flags=.....TC
48 2.237689 Intel_d1:b6:4f CiscoLinksys_f7:1d:... 802.11 54 QoS Null function (No data), SN=1487, FN=0, Flags=...P...TC
51 2.300697 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
52 2.302191 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=....R...C, BI=100, SSID="30 Munroe St"
53 2.304063 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=....R...C, BI=100, SSID="30 Munroe St"
54 2.305562 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=....R...C, BI=100, SSID="30 Munroe St"
55 2.308563 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=....R...C, BI=100, SSID="30 Munroe St"
56 2.310072 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2878, FN=0, Flags=....R...C, BI=100, SSID="30 Munroe St"
57 2.338148 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2879, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
58 2.440572 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2880, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
59 2.453941 CiscoLinksys_f7:1d:... Intel_1f:57:13 802.11 177 Probe Response, SN=2881, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
60 2.542945 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2882, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
61 2.645319 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2883, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
62 2.747697 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2884, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
63 2.850114 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2885, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
64 2.952572 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2886, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
65 3.054945 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2887, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
66 3.157343 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2888, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
67 3.260366 CiscoLinksys_f7:1d:... Broadcast 802.11 183 Beacon frame, SN=2889, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
68 3.260500 Intel_d1:b6:4f CiscoLinksys_f7:1d:... 802.11 54 QoS Null function (No data), SN=1488, FN=0, Flags=.....TC
Frame 52: Packet, 177 bytes on wire (1416 bits), 177 bytes captured (1416 bits)
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 Probe Response, Flags: ....R...C
IEEE 802.11 Wireless Management
Fixed parameters (12 bytes)
Timestamp: 174321321451
Beacon Interval: 0.102400 [Seconds]
Capabilities Information: 0x0601
....1 = ESS capabilities: Transmitter is an AP
....0. = IBSS status: Transmitter belongs to a BSS
.... .0.. = Reserved: 0
.... 0... = Reserved: 0
....0 .... = Privacy: Data confidentiality not required
....0. .... = Short Preamble: Not Allowed
.... .0.. .... = Critical Update Flag: False
.... 0... .... = Nontransmitted BSSIDs Critical Update Flag: False
....0 .... = Spectrum Management: Not Implemented
.... ..1. .... = QoS: Implemented
.... .1.. .... = Short Slot Time: In use
.... 0... .... = Automatic Power Save Delivery: Not Implemented
0000 00 00 18 00 ee 58 00 00 10 02 85 09 a0 00 e2 9c
0010 5e 00 00 46 81 c3 0d 1f 50 08 3a 01 00 12 f0 1f
0020 57 13 00 16 b6 f7 1d 51 00 16 b6 f7 1d 51 e0 b3
0030 eb 45 5c 96 28 00 00 00 64 00 01 06 00 0c 33 30
0040 20 4d 75 6e 72 6f 65 20 53 74 01 04 82 84 8b 96
0050 03 01 06 07 06 55 53 49 01 0b 1a 0c 12 0f 00 03
0060 a4 00 00 27 a4 00 00 42 43 5e 00 62 32 2f 00 2a
0070 01 00 32 08 8c 12 98 24 b0 48 60 6c dd 15 00 0a
0080 f5 0a 02 40 c0 00 03 01 03 05 0e 04 ff 00 03 00
0090 11 01 01 dd 18 00 50 f2 02 01 01 0f 00 03 a4 00
00a0 00 27 a4 00 00 42 43 5e 00 62 32 2f 00 81 c3 0d
00b0 1f
Wireshark_802_11.pcap
Packets: 2364 · Displayed: 1728 (73.1%)




```
┌──(kali㉿kali)-[~/Downloads/exp16]
└─$ wireshark Wireshark_802_11.pcap
^C

┌──(kali㉿kali)-[~/Downloads/exp16]
└─$ airodump-ng -r Wireshark_802_11.pcap

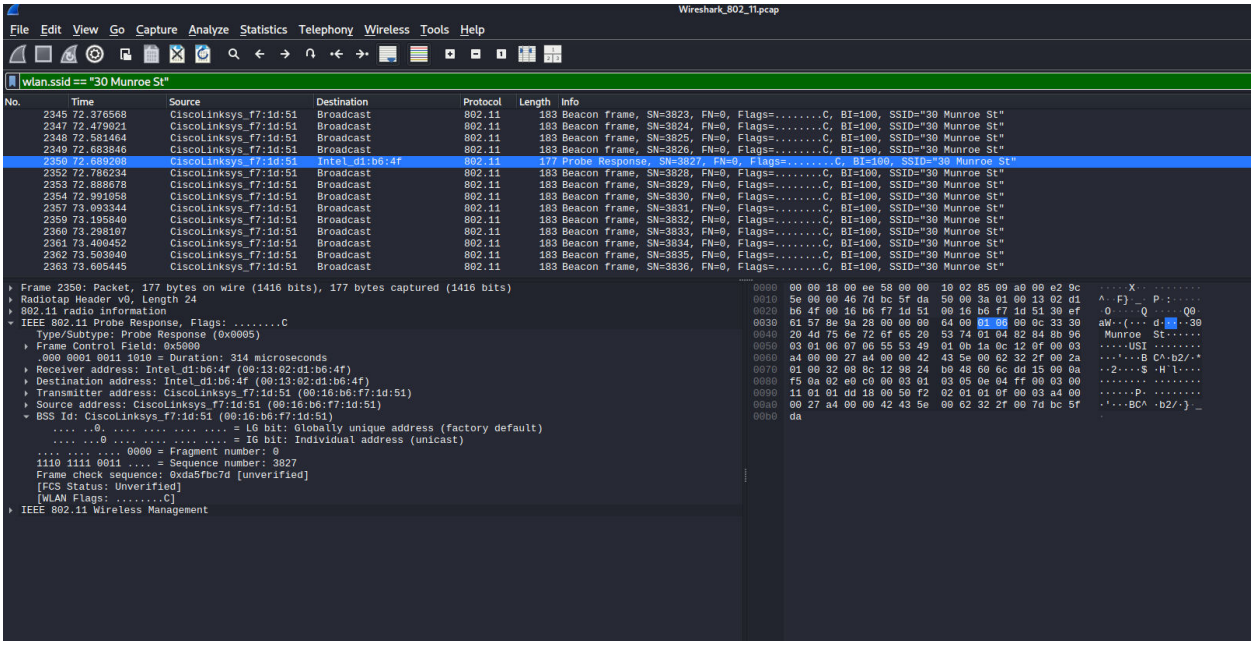
 CH  0 ][ Elapsed: 1 min ][ 2026-10-13 04:17 ][ Finished reading input file Wireshark_802_11.pcap.

 BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

 00:18:39:F5:BA:BB  -91        6       61    0   6   11   WPA  TKIP   PSK  linksys_SES_24086
 00:06:25:67:22:94  -92       32        0    0   6   11   WEP  WEP         linksys12
 00:16:B6:F7:1D:51  -28      718      479    0   6   54e  OPN              30 Munroe St

 BSSID              STATION            PWR    Rate    Lost   Frames  Notes  Probes

 (not associated)   00:12:F0:1F:57:13  -81    0 - 1      0        9         Home WIFI,phoiphas,concourse,linksys,hfmpc,BOHO2,BOWDOIN
 00:06:25:67:22:94  66:05:25:67:22:94  -92    0 - 2      0        1
 00:06:25:67:22:94  00:86:BC:D2:22:94  -93    0 - 2      0        1
 00:16:B6:F7:1D:51  00:13:02:D1:B6:4F  -26    1e-24e     0      799         30 Munroe St,linksys_SES_24086
```



RESULTS :

S. No.	Observation	Result
1	Capture file analysed	Wireshark_802_11.pcap
2	Capture type	IEEE 802.11 with Radiotap header
3	Number of packets	2,364
4	Wireless network identified	30 Munroe St
5	BSSID verified in Wireshark	00:16:b6:f7:10:51
6	Operating channel	6
7	Beacon interval	102.4 ms
8	Network privacy indicator	Data confidentiality not required
9	Client identified	Intel device, 00:13:02:d1:b6:4f
10	Tools used	Wireshark, TShark, Airodump-ng

CONCLUSION :

The wireless network footprinting experiment was performed successfully using an offline IEEE 802.11 packet capture. Wireshark was used to examine beacon and probe-response frames and identify the SSID, BSSID, channel, beacon interval, and privacy capability. Client MAC-address information was also observed, and Airodump-ng was used to inspect the detected wireless networks and stations. The experiment demonstrated how wireless network details can be identified from captured frames without performing a live scan.